

AADHAV S. BHARADWAJ

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US Citizen | GitHub | LinkedIn | Portfolio

EDUCATION

Case Western Reserve University

Aug 2023 – May 2027

B.S. Computer Science; Secondary Major in Mathematics; Minor in Economics

Cleveland, OH

GPA: 3.9/4.0 • Dean's High Honors List • Merit Scholarship

Relevant coursework: Machine Learning, Artificial Intelligence, Algorithms, Database Systems, Operating Systems, Efficient Deep Learning, Computer Security, Linear Algebra, Statistics

Minor in Economics: Microeconomics, Macroeconomics

London School of Economics and Political Science (LSE)

Summer 2025

Study Abroad — Economics

London, UK

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Java, SQL, R

AI/ML: RAG, Embeddings, LLM-as-Judge Evals, CI Eval Gates, Hybrid Retrieval (BM25 + RRF), Elasticsearch, pgvector, PyTorch

LLM Infrastructure: Langfuse, LangGraph, vLLM, AWS Bedrock, FastAPI, OpenAI-Compatible Gateways

Full Stack: Next.js, React, Node.js, Express, MongoDB, PostgreSQL

Real-Time / Infra: Apache Spark (PySpark), Socket.IO, WebRTC, mediasoup, Docker, CI/CD, AWS (Lambda, S3), Linux

EXPERIENCE

Abeyon

May 2026 – Present

Software Engineer Intern – AI/LLM

Washington, DC

- Implemented BM25 + dense hybrid retrieval with RRF in Elasticsearch, improving Recall@40 0.875→0.887 and MRR 0.555→0.610 on labeled evaluation sets.
- Reduced work-item generation latency from 123.6s to 49.7s (2.49×) via parallel NDJSON batch processing and migrated inference routing to AWS Bedrock across three applications.
- Built LLM-as-judge evaluation pipelines with frozen gold sets; diagnosed gate regression (recall 47%→98%), lifting E2E accuracy +7.2 pp.
- Deployed self-hosted Langfuse v3 and instrumented a production FastAPI LLM gateway (vLLM + AWS Bedrock) with tracing, custom pricing across 9 models, and structured logging for company-wide cost, token, and latency attribution.

Eaton Corporation

Sep 2025 – Jan 2026

AI Software Engineer Intern (contracted through xLab)

Remote

- Architected GenAI RAG chatbots for customer-interaction training using a MERN backend with vector retrieval and embedding search, improving onboarding efficiency by approximately 40%.
- Deployed containerized MERN platform with CI/CD, JWT/RBAC, and live WebSocket KPI dashboards for production trainer session analytics across modular training workflows.

WSOM, Case Western Reserve University

Jun 2025 – Present

Software Engineer Intern (AI Avatar Kiosk)

Cleveland, OH

- Built a dual-screen AI avatar kiosk using Next.js, AWS Lambda, HeyGen, and Pinecone-backed RAG; achieved sub-1.5s latency while serving 100+ daily users through CWRU SSO.

CWRU Computer Science Department

Aug 2024 – Present

Undergraduate Teaching Assistant

Cleveland, OH

- Lead weekly CS labs for 100+ students, hold 5+ office hours weekly, and developed Python autograding utilities supporting course operations.

PROJECTS

Blackbox-QA | Python, FastAPI, pgvector, Langfuse, LangGraph, Docker

- Built a framework-free tool-calling agent (bounded ReAct loop, 3 tools, read-only SQL guard) over hybrid retrieval (BM25 + dense + RRF, cross-encoder rerank) across a 30,646-report / 125,213-chunk corpus; lifted Recall@5 from 0.04 (BM25-only) to 0.64 and MRR to 0.62.
- Instrumented self-hosted Langfuse tracing with LLM-as-judge scoring via the Scores API; measured citation_match of 0.68 and answer_quality of 0.76 on a frozen gold set.
- Gated a query-rewrite retry on a calibrated cross-encoder score (TPR 1.0 / FPR 0.05) and added a deterministic Recall@5 CI regression gate on every pull request.
- Re-expressed the agent as an opt-in LangGraph multi-agent graph (retriever, SQL, critic, answer) with human-in-the-loop SQL approval via `interrupt()`.

Arguably | mediasoup, WebRTC, Socket.IO, Next.js

- Implemented a standalone mediasoup SFU and Socket.IO signaling stack (2.5k LOC) and integrated browser WebRTC functionality through custom media components.
- Added SFU unit testing, CI automation, HTTPS/CORS/TURN infrastructure, and production reliability improvements.

AWARDS

HackCWRU 2025 — 4th Place Overall, ExpressInk (AI Drawing Analysis)